

Bridging the Rust Reverse Engineering Gap: Automated Demangling and Function Identification in Ghidra

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Abstract

The growing adoption of the Rust programming language in both legitimate systems software and malware development has exposed critical limitations in existing reverse engineering frameworks. Rust binaries exhibit unique characteristics — aggressive monomorphization, static linking of the entire standard library, a specialized symbol mangling scheme (RFC 2603), and ownership-driven memory layout optimizations — that make them substantially more difficult to analyze than equivalent C/C++ programs. A controlled experiment by Microsoft Threat Intelligence Center demonstrated that a simple downloader program compiled in C++ produces fewer than 100 functions in a binary under 20 KB, while the same program in Rust yields nearly 10,000 functions in a binary exceeding 3 MB. This paper presents a method for automatically handling Rust executables within the Ghidra reverse engineering framework through the development of a specialized plugin. The plugin integrates four key capabilities: automated Rust binary detection via structural and string-based heuristics, full Rust v0 symbol demangling, standard library function identification using Ghidra's FunctionID database infrastructure, and Rust-specific data type annotation for structures such as `Option<T>`, `Result<T, E>`, string slices, and `Vec<T>`. We review the current state of Rust reverse engineering tooling — including GhidRust, ReOxide, the 0xA11C initiative, and Microsoft's RIFT — and position our contribution within this evolving landscape. Experimental evaluation on a corpus of 35 Rust binaries, including real-world malware samples such as BlackCat/ALPHV variants, demonstrates significant improvements in decompilation readability and a 40–60% reduction in manual analysis time.

Keywords: Rust, Ghidra, reverse engineering, decompilation, binary analysis, plugin, name demangling, FunctionID, malware analysis, static analysis, BlackCat, ALPHV.

1. Introduction

Reverse engineering of compiled binaries is a fundamental discipline within cybersecurity, spanning vulnerability research, malware analysis, software auditing, and digital forensics. Over the past decade, the complexity of this task has grown considerably as software ecosystems have diversified beyond the traditional C/C++ compilation model. The emergence of memory-safe systems languages — most notably Rust and Go — has created new categories of binary artifacts that existing analysis tools were not designed to handle.

The National Security Agency's Ghidra framework, released as open-source software under the Apache 2.0 license in March 2019, has become one of the most widely used platforms for binary analysis [1]. Ghidra offers a sophisticated decompiler engine based on the P-Code intermediate representation, an extensible plugin architecture supporting Java and Python scripting, and broad processor architecture coverage spanning x86, ARM, MIPS, PowerPC, and others. Its open-source nature has fostered a vibrant community of contributors and extension developers, making it the primary free alternative to the commercial IDA Pro disassembler [2].

Concurrently, the Rust programming language has experienced rapid growth across domains where memory safety, performance, and concurrency are critical [3]. Rust's ownership

model and affine type system eliminate entire classes of memory corruption vulnerabilities — use-after-free, double-free, buffer overflows, and data races — at compile time without requiring a garbage collector. These properties have attracted adoption in operating system kernels (the Linux kernel began accepting Rust code in version 6.1), web browsers (Mozilla Firefox’s Servo engine), blockchain implementations, and embedded systems.

However, the same properties that make Rust appealing to legitimate developers have also attracted malware authors. ALPHV/BlackCat, first observed in late 2021, became the first professionally deployed ransomware written in Rust, operating as a Ransomware-as-a-Service (RaaS) platform [4]. The Hive ransomware group rewrote their payload in Rust in mid-2022 to improve cross-platform support and evade detection [5]. More recently, RALord ransomware, the SPICA backdoor attributed to nation-state actors, and updated variants of the Buer downloader (dubbed “RustyBuer”) have demonstrated the accelerating trend toward Rust-based malware [6, 7]. As Juan Andrés Guerrero-Saade of SentinelLabs stated at Black Hat 2024, with the current state of existing tooling, Rust remains exceptionally challenging to reverse engineer, causing many analysts to avoid the Rust malware ecosystem entirely [8].

The core difficulty stems from fundamental differences between Rust and C/C++ compilation models. Microsoft Threat Intelligence Center demonstrated this gap with a controlled experiment: a simple PE executable that downloads a file and saves it to disk, when compiled in C++ with MSVC in release mode, produces a dynamically linked binary with fewer than 100 functions and a size under 20 KB. The identical program compiled in Rust with default release settings generates a statically linked binary containing approximately 10,000 functions and exceeding 3 MB [6]. This hundred-fold increase in function count arises because Rust statically links the standard library, allocator, panic infrastructure, formatting machinery, and all dependency crates into a single monolithic binary.

This paper presents a method for automatically handling Rust executables in Ghidra through the development of a plugin that automates key analysis and decompilation tasks. The contributions of this work are: (1) a systematic review of Rust binary analysis challenges grounded in current research and real-world tool analysis; (2) an automated multi-heuristic detection mechanism for identifying Rust binaries; (3) a complete implementation of Rust v0 symbol demangling integrated into Ghidra’s analysis pipeline; (4) a FunctionID-based approach for annotating standard library functions across multiple compiler versions; (5) heuristics for reconstructing Rust-specific data types in decompiled output; and (6) an empirical evaluation demonstrating the effectiveness of the approach on a diverse corpus including malware samples.

2. Background

2.1. Ghidra Architecture and Extension Model

Ghidra’s architecture follows a dual-process design: a Java-based front end handles user interaction, program representation, and the plugin ecosystem, while a separate C++ program implements the decompiler engine [1]. Communication between the two occurs through a well-defined interface that passes function metadata, data types, and decompiled output.

The decompiler operates on P-Code, a register transfer language (RTL) that abstracts processor-specific machine code into a uniform intermediate representation. The SLEIGH specification language governs the translation of machine instructions into P-Code operations. Each processor architecture has a corresponding SLEIGH specification that defines instruction decoding and P-Code emission rules [9]. After translation, the decompiler applies a pipeline of transformations: SSA (Static Single Assignment) construction with incremental phi-node placement, constant propagation, dead code elimination, copy propagation, type inference, and finally C-like output emission [10].

Ghidra supports several extension mechanisms relevant to this work. Analyzer plugins register into the auto-analysis pipeline at configurable priority levels, executing during the automated analysis phase that follows binary import. The FunctionID subsystem provides hash-based function identification: it computes an operand-masked hash over a function's body (preserving opcodes and prefixes while zeroing register, immediate, and displacement operands) and matches the resulting fingerprint against pre-built .fidb databases [11]. This mechanism was originally designed for C/C++ library identification but can be applied to any statically linked code, including Rust standard library functions.

Critically, the decompiler's C++ codebase is not directly extensible through the standard Java API. This means that certain analysis improvements — such as modifying the decompiler's type inference rules or adding Rust-specific simplification passes — require either modifications to the decompiler source code or specialized frameworks like ReOxide [12]. Our plugin operates within the standard Java API boundary, applying its improvements through the symbol table, function signatures, and data type definitions that the decompiler consults during its output generation phase.

2.2. Rust Compilation Pipeline and Binary Characteristics

Understanding the Rust compilation pipeline is essential for effective reverse engineering. The Rust compiler (rustc) processes source code through several intermediate stages before producing machine code [3]. The initial parsing and macro expansion phase produces an Abstract Syntax Tree (AST). Type checking, trait resolution, and borrow checking occur on the High-level Intermediate Representation (HIR). The code is then lowered to Mid-level Intermediate Representation (MIR), where ownership and borrowing rules are formally verified and initial optimizations are applied. Finally, MIR is translated to LLVM IR, which undergoes LLVM's optimization pipeline before machine code emission.

Several aspects of this pipeline produce binary characteristics that challenge reverse engineering tools designed for C/C++ binaries.

2.2.1. Monomorphization and Code Bloat

Rust's generics are implemented through monomorphization: for each concrete type instantiation of a generic function, the compiler generates a separate, specialized copy of the machine code. A generic function like `Vec<T>::push` generates independent implementations for `Vec<u32>`, `Vec<String>`, `Vec<Box<dyn Error>>`, and every other type with which it is instantiated across the entire dependency tree. While this strategy enables zero-cost abstractions (no runtime dispatch overhead), it dramatically increases binary size and the number of functions that must be analyzed.

The impact of monomorphization is compounded by Rust's ecosystem culture, which favors small, composable libraries (crates) with extensive use of generics. A typical web server application might depend on dozens of crates, each contributing monomorphized function instantiations. The result is a binary where the actual application logic constitutes a small fraction of the total code, with the remainder consisting of library function copies that are, in stripped binaries, indistinguishable from application code.

2.2.2. Static Linking

Unlike C/C++ programs, which typically link dynamically against system libraries (libc, libstdc++), Rust binaries are statically linked by default. The Rust standard library (std), the core library (core), the allocation library (alloc), and all dependency crates are compiled and linked directly into the final binary. This produces self-contained executables with no external library dependencies beyond the system's dynamic linker and libc (on Linux) or the Windows API (on Windows).

For reverse engineering, static linking eliminates the helpful boundary that dynamic linking provides between application code and library code. In a dynamically linked C program, calls to

printf or malloc are immediately identifiable through the PLT/GOT mechanism. In a statically linked Rust binary, the equivalent functionality is embedded within the binary and, when stripped of symbols, appears as ordinary unnamed functions.

2.2.3. Symbol Mangling

Rust uses a custom symbol mangling scheme that encodes rich type and path information into function names. The v0 scheme, defined in RFC 2603 and implemented since Rust 1.37 (2019), uses the prefix “_R” followed by a structured encoding of crate disambiguators, namespace tags (value “v”, type “t”, closure “C”, shim “S”), identifier lengths with optional Punycode encoding for Unicode names, and recursively encoded generic argument lists [13]. The earlier legacy mangling format uses the “_ZN” prefix shared with C++ Itanium mangling but with Rust-specific path structures and a trailing hash.

The v0 scheme was designed with demangling efficiency in mind: the encoding can be parsed in a single pass without backtracking or complex data structures [13]. However, the raw mangled names are effectively unreadable. For example, the mangled symbol “_RNVtCsaBcDeFgHiJ_7mycrate4util10parse_data” represents the function `mycrate::util::parse_data` with a crate disambiguator hash. Without demangling, an analyst confronting a stripped binary sees only these cryptic identifiers, if any symbols remain at all.

2.2.4. Data Type Layout Optimizations

Rust’s type system includes several constructs that compile to non-obvious memory layouts. Enum types — Rust’s algebraic data types — compile to tagged unions with a discriminant field indicating the active variant. The compiler applies niche optimization: when one variant contains a type with invalid bit patterns (such as a non-null reference), the compiler uses those invalid values to represent other variants, eliminating the discriminant entirely [14].

The most common example is `Option<T>`, where the `None` variant is represented by the null pointer value (0x0) and `Some(&T)` by the non-null pointer itself. This means `Option<T>` has the same size as a raw pointer, with no discriminant overhead. The same optimization applies to `Option<Box<T>>`, `Option<NonNull<T>>`, and other non-nullable types. For the reverse engineer, this means that what appears to be a simple null check in the disassembly is actually a Rust `Option::is_none()` test, and understanding this distinction is crucial for correctly interpreting the program’s logic.

The `Result<T, E>` type, which is Rust’s primary error handling mechanism, undergoes similar optimizations. Each use of the “?” operator compiles to a discriminant check followed by either unwrapping the `Ok` value for continued execution or early return with the `Err` value, often passing through a `From` trait conversion. These patterns generate distinctive control flow structures: a branch on a tag value, one path continuing normally and the other performing a return with value transformation.

String slices (`&str`) and array slices (`&[T]`) are represented as fat pointers: a (pointer, length) pair occupying 16 bytes on 64-bit platforms. The `Vec<T>` collection type extends this to a (pointer, length, capacity) triple of 24 bytes. These multi-field representations appear as anonymous structures in decompiled output unless the reverse engineering tool provides Rust-aware type definitions.

2.2.5. Panic and Unwinding Infrastructure

Rust’s panic system, which handles unrecoverable errors, introduces substantial code into the binary even when the application itself never panics. The default panic strategy (unwinding) includes stack unwinding tables, personality functions, and landing pads similar to C++ exception handling. Calls to functions like `core::panicking::panic`, `core::panicking::panic_fmt`, and `std::process::abort` are scattered throughout the binary as the compiler inserts them at every point where a runtime check can fail (array bounds checks, arithmetic overflow in debug mode, `unwrap()` calls on `None/Err` values). Identifying these panic paths is important for the analyst

because they represent error-handling infrastructure rather than application logic, yet in a stripped binary they consume significant analytical attention.

2.3. The Rust Malware Landscape

The adoption of Rust by malware authors has accelerated markedly since 2021, creating urgency for improved reverse engineering capabilities.

ALPHV/BlackCat, first documented by Recorded Future in December 2021, was the first professionally operated ransomware written in Rust [4]. Operating as a Ransomware-as-a-Service (RaaS) platform, BlackCat was advertised on underground cybercrime forums (XSS and Exploit), offering affiliates 80–90% of ransom payments. The ransomware supported Windows, Linux, and VMware ESXi targets, demonstrating Rust’s cross-compilation capabilities. Technical analysis revealed sophisticated use of Rust crates: Zeroize for secure memory wiping, AES and ChaCha20 for encryption, and command-line argument parsing for runtime configuration [15]. BlackCat required a 32-character access token for execution, used named pipes for inter-process communication, and implemented UAC bypass and privilege escalation. By late January 2022, the group’s leak site had named over twenty victim organizations [4].

The Hive ransomware group migrated their payload from Go to Rust in June 2022, citing improved encryption performance and better evasion of detection signatures [5]. This migration illustrates a broader trend: malware authors adopt Rust not merely for its technical merits but also because the resulting binaries are harder for security tools to analyze, effectively providing “obfuscation through compilation.” More recently, RALord ransomware and the SPICA backdoor (attributed to a nation-state actor) have been analyzed using RIFT, Microsoft’s specialized Rust analysis tool [6].

The cybersecurity community has responded with several initiatives. The 0xA11C project, launched at Black Hat 2024 by SentinelLabs and Intezer, aims to develop methodologies and open-source tools specifically for Rust malware reverse engineering [8]. This initiative builds on the AlphaGolang approach that SentinelLabs previously developed to address the similar challenges posed by Go-compiled malware. Patrick Ventuzelo’s Ringzer0 training course on Rust reverse engineering, offered at BOOTSTRAP25, provides hands-on instruction covering tools, data structures, obfuscation, and malware techniques [16].

2.4. Existing Tools and Their Limitations

Several tools address aspects of Rust binary analysis, but none provide a comprehensive, automated solution within the Ghidra framework.

GhidRust is a Ghidra extension developed as an academic project that implements three capabilities: Rust binary detection, standard library function identification via FunctionID (.fidb) databases (with a pre-built database for x86-64 Rust 1.58.1), and a prototype C-to-Rust transpiler for decompiled output [17]. However, development has been permanently paused. The author acknowledges that Ghidra’s native Rust identification has since surpassed the plugin’s detection capabilities, and the remaining features (FunctionID integration and transpilation) are described as “quite flaky.” The project nonetheless established the viability of FunctionID-based Rust library identification and provided a template for subsequent work.

ReOxide, developed by SBA Research with NLnet/NGIO Entrust funding through the European Commission’s Next Generation Internet program, takes a fundamentally different approach [12]. Rather than operating through Ghidra’s Java API, ReOxide adds a plugin system directly to the Ghidra decompiler’s C++ codebase. This deep integration allows modification of the decompiler’s internal analysis passes — information about variable liveness, type constraints, and control flow that is available during decompilation but not exposed through the public API. ReOxide distributes pre-compiled decompiler binaries via a Python pip package and supports both C++ and Rust plugin development. While architecturally sophisticated, ReOxide remains in active development and is not yet suitable for turnkey deployment.

RIFT (Rust Library Recognition Project), released by Microsoft Threat Intelligence Center in June 2025 and presented at RECON 2025, represents the most complete tool currently available [6]. RIFT consists of three components: a Static Analyzer (IDA Pro plugin) that extracts Rust compiler commit hashes and dependency metadata from binaries; a Generator (Python script) that automates FLIRT signature generation and binary diffing using Diaphora; and a Diff Applier (IDA Pro plugin) that applies annotations. RIFT combines two complementary techniques: FLIRT signatures provide fast, high-precision matching for unmodified library functions, while binary diffing handles functions that have been slightly altered by different compilation settings. Microsoft validated RIFT on RALord ransomware and the SPICA backdoor, demonstrating effective separation of library code from malicious logic. However, RIFT is exclusively designed for IDA Pro (versions 9.0+) and provides no Ghidra support.

Ghidra 11.0 native Rust support (released January 2024) introduced two features: improved function and variable name demangling with RustPath awareness for parsing Rust-specific constructs such as traits and generic instantiations, and a new RustStringAnalyzer [18]. The RustStringAnalyzer identifies Rust string slice references in the binary's read-only data sections, creating individual data types for each substring reference rather than treating them as offsets into a monolithic string table. This significantly improves both the decompiler output readability and the quality of cross-reference analysis. Nevertheless, these additions address only a subset of the challenges: standard library identification, data type reconstruction, and Result/Option pattern recognition remain unaddressed.

Table 1 summarizes the comparative capabilities of existing tools and our proposed plugin.

Table 1. Comparative analysis of Rust reverse engineering tools. GhidRust supports Ghidra with binary detection, FunctionID, and basic transpilation but is discontinued. ReOxide extends the Ghidra decompiler at the C++ level but is still in development. RIFT offers FLIRT and binary diffing for IDA Pro only. Ghidra 11.0 provides native demangling and string analysis. Our proposed plugin targets Ghidra with detection, v0 demangling, FunctionID across multiple compiler versions, and data type reconstruction — filling the gap left by discontinued and platform-exclusive tools.

3. Proposed Method

The proposed method integrates four interconnected components within a unified Ghidra plugin implemented in Java: Rust binary detection, symbol demangling, standard library identification, and data type reconstruction. The components are designed to execute sequentially within Ghidra's auto-analysis pipeline, with each stage enriching the program database for subsequent stages and ultimately improving the decompiler's output quality.

3.1. Automated Rust Binary Detection

The detection module is the entry gate for all subsequent Rust-specific analysis. It employs a multi-heuristic scoring approach that combines evidence from several independent indicators to produce a confidence score.

3.1.1. String-Based Heuristics

The module scans the binary's read-only data sections (.rodata in ELF, .rdata in PE) for strings characteristically embedded by the Rust compiler and standard library. These include: (a) paths prefixed with `"/rustc/"`, which reference the Rust compiler's source tree and appear in panic messages generated by the standard library; (b) the `"RUST_BACKTRACE"` string, used by the Rust runtime for stack trace capture and present in virtually all Rust binaries; (c) format string patterns from the `core::fmt` module, which have a distinctive structure due to Rust's compile-time format string parsing; and (d) the `"called `Option::unwrap()` on a `None` value"` and similar panic messages that are specific to Rust's standard library [18].

3.1.2. Structural Heuristics

ELF binaries compiled by `rustc` contain a `".rustc"` section that stores compiler metadata, including version information. The presence of this section is a strong positive indicator. For PE

binaries, the module examines import table entries for functions associated with the Rust runtime, such as those from the “bcrypt” (for random number generation), “ws2_32” (networking), and “ntdll” libraries in patterns characteristic of Rust’s standard library rather than direct application use.

The entry point analysis examines the binary’s startup sequence for the lang_start function, which is the actual entry point for Rust programs. In a standard Rust binary, the operating system’s entry point calls a series of initialization functions that eventually invoke std::rt::lang_start, which in turn calls the user’s main function. This layered entry point structure differs from the direct main() call typical of C programs.

3.1.3. Confidence Scoring

Each heuristic contributes a weighted score: the .rustc section receives the highest weight (0.4) as the most reliable indicator, followed by /rustc/ path strings (0.25), RUST_BACKTRACE presence (0.15), entry point structure (0.1), and format string patterns (0.1). A binary exceeding a configurable threshold (default: 0.5) is classified as Rust-compiled. This weighted approach accommodates partial evidence: a stripped binary might lack the .rustc section but still be detectable through string patterns and entry point analysis.

3.2. Rust v0 Symbol Demangling

Symbol demangling is the most immediately impactful analysis step, transforming cryptic mangled names into human-readable qualified identifiers that reveal the binary’s module structure.

3.2.1. V0 Grammar Implementation

The plugin implements a recursive descent parser for the full v0 mangling grammar as specified in RFC 2603 [13]. The v0 encoding begins with the prefix “_R”, followed by a path encoding that recursively describes the symbol’s location within the Rust namespace hierarchy. The grammar supports several path productions: crate roots (tagged with “C”), inherent implementations (“M”), trait implementations (“X”), nested paths (“N” with namespace tags for values, types, closures, and compiler-generated shims), and generic instantiations (“I”).

Identifiers within paths are encoded as a decimal length prefix followed by the identifier bytes, with optional Punycode encoding (signaled by a “u” prefix) for non-ASCII Unicode identifiers. Disambiguator hashes use a base-62 encoding with the “s” prefix. Generic argument lists encode types recursively, supporting primitive types (single-character codes: “b” for bool, “c” for char, “h” for u8, “m” for u32, etc.), references (“R” for shared, “Q” for mutable), tuples (“T”), arrays (“A”), slices (“S”), and named types via recursive path encoding.

3.2.2. Legacy Format Fallback

The implementation also includes a fallback demangler for the pre-v0 legacy Rust mangling format. Legacy symbols use the “_ZN” prefix (shared with C++ Itanium mangling) followed by a sequence of length-prefixed path components and a trailing 17-character hash (one “h” character plus 16 hex digits). The fallback demangler strips the hash and reconstructs the path using “::” separators. While less information-rich than v0 demangling (generic parameters are not encoded), the legacy demangler ensures coverage of binaries compiled with Rust versions prior to 1.37.

3.2.3. Integration with Ghidra

Demangled names are applied through Ghidra’s SymbolTable and FunctionManager APIs. For each successfully demangled function, the plugin: (a) sets the function’s primary label to the demangled name; (b) creates namespace objects corresponding to the crate and module hierarchy (e.g., mycrate::util:: becomes a nested Namespace structure in Ghidra’s program tree); and (c) where generic instantiation information is available, adds plate comments documenting the concrete type parameters. This hierarchical organization allows analysts to collapse or filter library namespaces (std::, core::, alloc::) in Ghidra’s Symbol Tree view, immediately revealing the application-specific code.

3.3. Standard Library Function Identification

Identifying statically linked standard library functions is essential for separating application logic from library code. The plugin leverages Ghidra's FunctionID infrastructure for this purpose.

3.3.1. *FunctionID Mechanism*

Ghidra's FunctionID system computes a two-level hash for each function [11]. The "full hash" is computed over the function body with operand bytes masked (zeroed), capturing the instruction pattern while tolerating address-dependent variations. The "specific hash" combines the full hash with hashes of the function's direct call targets, adding structural context. A match requires agreement of both hashes, providing high precision with low false positive rates.

FunctionID databases (.fidb files) are self-contained SQLite databases that can be generated, attached, and shared among Ghidra users. Each database records function hashes along with their associated names, library identifiers, and version strings. During auto-analysis, the FunctionID analyzer computes hashes for unknown functions and queries attached databases for matches [11].

3.3.2. *Rust-Specific Database Generation*

We generate FunctionID databases through a systematic process. For each target compiler version (we cover Rust stable 1.60 through 1.82) and target triple (x86_64-unknown-linux-gnu and x86_64-pc-windows-msvc), we compile the Rust standard library with debug symbols using rustup component add rust-src and cargo build with appropriate flags. The resulting .rlib files (Rust's archive format containing COFF/ELF object files) are extracted and imported into Ghidra for analysis.

During the FunctionID ingestion phase, we follow Ghidra's recommended practices: the FunctionID analyzer itself is disabled to prevent cross-contamination between databases, and the Demangler analyzer is disabled to preserve raw mangled symbols, allowing the symbols to be applied correctly during future analysis passes [11]. The generated databases are packaged alongside the plugin distribution.

3.3.3. *Version Sensitivity Challenge*

A fundamental limitation of hash-based function identification is its sensitivity to compilation variations. Even minor differences in compiler version, optimization level, or target features can alter instruction selection, register allocation, or basic block ordering, causing hash mismatches. This is a known challenge: Ghidra's documentation notes that functions compiled with different optimization levels will not match, and the compiler version is especially important for Rust binaries [11].

Our database spanning 22 compiler versions provides reasonable coverage, but analysts encountering binaries compiled with unrepresented versions will experience reduced match rates. To address this, the plugin provides a utility script for generating new databases from reference compilations, and we document the process for analysts to extend coverage for their specific needs. The binary diffing approach used by Microsoft's RIFT [6] represents a complementary technique that trades analysis time for version tolerance; integrating a similar approach within Ghidra is a direction for future work.

3.4. Rust Data Type Reconstruction

The final analysis stage applies Rust-aware data type definitions to improve decompiler output readability.

3.4.1. *String Slices and Fat Pointers*

Rust string slices (&str) and byte slices (&[u8]) are represented as fat pointers consisting of a data pointer and a usize length field, occupying 16 bytes on x86-64. The plugin defines a custom Ghidra structure type RustStrSlice with fields {data_ptr: pointer, len: usize} and applies it where the following conditions are met: (a) two consecutive 8-byte values are used together in function

calls identified as string operations (e.g., `core::fmt::write` or `std::io::Write::write_all`); and (b) the first value is used as a pointer operand (load/store) and the second as a length parameter.

3.4.2. *Option<T> and Niche Optimization*

For Option types with niche optimization, the plugin annotates null-pointer checks as `Option::is_none()` / `Option::is_some()` predicates. The detection heuristic identifies code patterns where: (a) a pointer value is tested against zero; (b) the zero-test branch leads to either a function return or a panic call (unwrap on None); and (c) the non-zero branch uses the pointer value as a valid reference. These patterns are annotated with plate comments in the decompiler output, such as “// Rust: Option::is_none() check — None branch below.”

3.4.3. *Result<T, E> and Error Propagation*

Result type patterns are identified by recognizing discriminant-based branching structures: a load of a tag value (typically the first byte or word of a structure), a comparison against zero or one, and divergent paths where one continues execution and the other performs a return with value transformation (the “?” operator pattern). When the error-return path passes through functions identified as `From::from` trait implementations (via demangling), the entire pattern is annotated as a Result propagation.

3.4.4. *Vec<T> and Collections*

`Vec<T>` is modeled as a three-field structure {buf: pointer, len: usize, cap: usize} of 24 bytes. The plugin identifies Vec instances where allocation and growth functions (`alloc::raw_vec::RawVec<T, A>::grow_one`, `alloc::raw_vec::finish_grow`) are detected in the call graph through standard library identification. The corresponding local variables are retyped to the `RustVec` structure, and heap-related operations (`alloc`, `realloc`, `dealloc`) are annotated with comments indicating their role in Vec lifecycle management.

4. Implementation

4.1. Architecture

The plugin is implemented in Java 17 as a Ghidra extension module comprising approximately 9,200 lines of code organized into five packages: detect (binary identification heuristics), demangle (v0 and legacy symbol processing), fid (FunctionID database management and matching), types (data structure pattern matching and type application), and ui (user interface extensions).

The plugin registers itself as a Ghidra Analyzer with the name “Rust Binary Analyzer” and a priority level of `AnalysisPriority.DATA_TYPE_PROPOGATION.before()`. This priority ensures execution after the disassembler, basic block model, and reference analysis have completed, but before the decompiler’s parameter identification pass. This ordering is critical: the plugin’s type annotations and corrected function signatures must be present in the program database before the decompiler reads them, as the decompiler is invoked on a per-function basis during later analysis phases.

4.2. Demangling Engine

The demangling component is implemented as a standalone recursive descent parser (~1,400 lines of Java). The parser operates in two modes, inspired by the LLVM RustDemangle implementation [19]: a “skip” mode that validates the encoding structure without producing output, and a “print” mode that constructs the demangled string via a `StringBuilder`. This dual-mode design allows efficient validation of deeply nested symbols before committing to output generation.

The parser maintains a recursion depth counter with a configurable maximum (default: 128) to prevent stack overflow on pathological or malformed inputs. Punycode decoding is implemented with a fixed 256-byte stack buffer sufficient for identifiers up to 256 characters; longer identifiers fall back to heap allocation. The parser handles all v0 grammar productions

including const generics (introduced in Rust 1.51), associated type bindings, higher-ranked trait bounds with bound lifetime depth tracking, and compiler-generated shim functions.

Performance testing on a binary containing 12,847 mangled symbols showed a complete demangling pass executing in approximately 85 milliseconds on a modern workstation (Intel i7-12700, 32 GB RAM), confirming that the demangling engine does not introduce perceptible delay in the analysis pipeline.

4.3. FunctionID Database Pipeline

The FunctionID database generation pipeline is automated through a shell script that orchestrates the following steps for each (compiler_version, target_triple) pair: (1) install the specified Rust toolchain via rustup; (2) compile the standard library with debug symbols using cargo build --release with the rust-src component; (3) extract .rlib archives and decompress the contained object files; (4) import each object file into a headless Ghidra project using analyzeHeadless with the FunctionIDHeadlessPrescript.java script to disable interfering analyzers; (5) ingest functions into the .fidb database using Ghidra's FunctionID plugin; and (6) validate the database by spot-checking known function signatures.

The resulting database collection spans 22 Rust stable versions (1.60–1.82) across two target triples, containing approximately 47,000 unique function signatures per version. The total database package occupies approximately 180 MB in compressed form.

4.4. User Interface

The plugin extends Ghidra's tool windows with a "Rust Analysis" panel accessible from the Window menu. The panel displays: (a) detection results including the confidence score and individual heuristic contributions; (b) demangling statistics showing total symbols processed, v0 success rate, legacy fallback rate, and failure count; (c) FunctionID coverage showing the number of matched standard library functions, the matched compiler version, and estimated library-to-application code ratio; and (d) data type annotation counts.

A context menu integration on individual functions provides three actions: "Force Rust Demangle" (manually triggers demangling for a selected function), "Apply Rust Type" (opens a dialog for manually specifying a Rust type annotation), and "Mark as Library" (flags a function as library code for filtering purposes). The Symbol Tree view is enhanced with a filter toggle that hides functions in the std::, core::, and alloc:: namespaces, providing a decluttered view of application-specific code.

5. Case Study: Analysis of a Rust-Compiled Utility

To illustrate the plugin's practical impact, we present a case study analyzing a stripped Rust-compiled command-line utility (a JSON processing tool) with and without the plugin.

5.1. Baseline Analysis (Without Plugin)

The binary, compiled with Rust 1.75 in release mode and stripped of symbols, contains 8,247 functions identified by Ghidra's default analysis. All functions are labeled with auto-generated names (FUN_00xxxxxx). The decompiler output for the main processing function shows a 340-line C function with dozens of calls to unnamed functions, opaque 16-byte and 24-byte structure manipulations, and numerous branches that appear to be error handling but cannot be distinguished from application logic. An experienced analyst estimated that understanding this single function's purpose would require approximately 4–6 hours of manual analysis.

5.2. Analysis with Plugin

With the plugin enabled, the analysis proceeds as follows. The detection module identifies the binary as Rust-compiled with a confidence score of 0.85 (triggered by /rustc/ paths and RUST_BACKTRACE strings; the .rustc section was stripped). The demangling pass recovers 7,412 function names (89.9%), revealing the crate and module structure: the application's own logic resides in 127 functions within the jsonutil:: namespace, while the remaining 8,120 functions belong to std::, core::, alloc::, serde::, serde_json::, and clap:: namespaces.

The FunctionID pass matches 5,893 functions (71.4%) against the Rust 1.75 database entry. Combined with demangling, 97.3% of functions now have meaningful identifiers. The Symbol Tree filter hides library namespaces, reducing the visible function list to 127 application functions.

The data type reconstruction pass identifies 34 Option null checks, 18 Result discriminant patterns, and 45 string slice references. The decompiler output for the main processing function now shows calls to `serde_json::from_str`, `std::fs::read_to_string`, and `std::io::Write::write_all` with proper names, Option/Result annotations on branches, and string slices properly typed. The same analyst estimated that understanding the function now requires approximately 30–45 minutes — an 85–90% reduction in analysis time.

6. Experimental Evaluation

6.1. Setup

The plugin was evaluated on a corpus of 35 Rust binaries divided into three categories: (a) 12 open-source utilities compiled from source with known ground truth, spanning simple CLI tools to moderately complex network applications (1,500–18,000 functions); (b) 8 CTF challenge binaries from recent competitions (DEFCON, PlaidCTF, GoogleCTF 2023–2024) that use Rust for anti-reverse-engineering purposes; and (c) 15 real-world malware samples identified as Rust-compiled, including BlackCat/ALPHV variants, Hive payloads, and other ransomware families. All binaries targeted x86-64, with a mix of Linux ELF and Windows PE formats. Both debug and stripped release configurations were represented.

Evaluation was conducted with Ghidra 11.2 on Ubuntu 22.04 (Intel i7-12700, 32 GB RAM). Each binary was analyzed twice: once with Ghidra’s default analysis pipeline and once with the plugin enabled. Metrics included demangling coverage, FunctionID match rate, analysis time overhead, and qualitative decompilation readability assessed by two experienced reverse engineers on a 5-point Likert scale.

6.2. Quantitative Results

6.2.1. Detection Accuracy

The detection module correctly identified all 35 Rust binaries (100% recall) and produced no false positives on a control set of 20 non-Rust binaries compiled from C, C++, and Go (100% precision). Among the Rust binaries, 23 triggered the `.rustc` section heuristic, while the remaining 12 (stripped of section metadata) were identified through string patterns and entry point analysis alone, with confidence scores ranging from 0.52 to 0.78.

6.2.2. Demangling Coverage

Symbol demangling achieved an average success rate of 95.1% for v0-mangled symbols. The remaining 4.9% consisted of legacy-format symbols (3.1%), partially recovered by the fallback demangler to bring effective coverage to 96.8%, and non-conformant or truncated encodings (1.8%). Demangling performance varied by category: open-source utilities with intact symbol tables achieved 99.2% coverage, while stripped malware samples retained fewer symbols (average 2,340 per binary) but achieved similar per-symbol success rates (94.7%).

6.2.3. FunctionID Matching

FunctionID matching performance depended strongly on compiler version alignment. For binaries compiled with a version present in the database (28 of 35 samples), the average match rate was 76.8% of total functions and 89.2% of standard library functions specifically. For the 7 binaries compiled with unrepresented versions (including nightly builds), match rates dropped to 48.3%. Across the full corpus, the weighted average was 74.2% for stripped release binaries and 93.7% when debug symbols supplemented identification.

6.2.4. Analysis Time

The plugin introduced an average overhead of 18.5% over baseline Ghidra analysis time. The breakdown by component was: detection (0.3%), demangling (2.1%), FunctionID matching (14.8%), and data type reconstruction (1.3%). For a medium-complexity binary (~6,000 functions),

total analysis time averaged 3 minutes 50 seconds versus 3 minutes 15 seconds for baseline. The FunctionID component dominates overhead due to hash computation for all unknown functions; this could be optimized through parallel hash computation or incremental matching.

6.3. Qualitative Assessment

Two reverse engineers independently assessed decompilation readability on a randomly selected subset of 10 functions from each binary category (30 functions total). Mean readability scores improved from 2.2 (baseline Ghidra, $\sigma=0.7$) to 3.9 (with plugin, $\sigma=0.5$) on the 5-point Likert scale, representing a statistically significant improvement ($p < 0.001$, paired t-test, $n=60$).

Reviewers identified three primary sources of improvement: (1) the ability to immediately distinguish application functions from library functions reduced “needle in haystack” effort; (2) demangled function names at call sites provided contextual understanding without requiring traversal into called functions; and (3) data type annotations on Option/Result patterns reduced misinterpretation of error-handling branches as application logic. The primary remaining complaint was that the decompiler still produces C-like output rather than Rust syntax, which requires ongoing mental translation for Rust-specific constructs.

7. Discussion

7.1. Positioning Among Existing Tools

Our plugin occupies a specific niche in the Rust reverse engineering tool landscape. It provides a turnkey, automated analysis improvement for Ghidra users — a category that includes many analysts who work with Ghidra rather than IDA Pro due to cost, licensing, or organizational constraints. Unlike RIFT, which requires IDA Pro 9.0+ and its commercial license, our plugin runs on the freely available Ghidra platform. Unlike ReOxide, which requires building from source and learning a new plugin development API, our plugin installs as a standard Ghidra extension with immediate functionality.

The tools are complementary rather than competitive. An analyst with access to both IDA Pro and Ghidra might use RIFT for FLIRT-based library identification and our plugin for Ghidra-specific features such as the P-Code-based decompiler integration and namespace hierarchy construction. ReOxide’s decompiler-level modifications represent a deeper form of analysis improvement that our plugin’s Java API-level approach cannot replicate; as ReOxide matures, integration points between the two systems could yield compound benefits.

7.2. Limitations

The signature-based FunctionID approach is fundamentally version-sensitive. Our evaluation showed a 28.5-percentage-point drop in match rates for binaries compiled with unrepresented compiler versions. Microsoft’s RIFT addresses this with binary diffing, which tolerates compilation variations at the cost of substantially longer analysis time (hours vs. seconds) [6]. Integrating a lightweight binary similarity metric, potentially based on Ghidra’s BSim (Binary Similarity) feature, could provide a middle ground between strict hash matching and full binary diffing.

Data type reconstruction remains heuristic and pattern-based. Complex enum layouts with multi-variant niche filling, as produced by recent compiler versions, are not reliably detected. The plugin’s Option and Result identification works well for common patterns but may miss unusual generic instantiations or custom error types with non-standard layouts.

The plugin currently supports x86-64 exclusively. Extension to AArch64 (critical for mobile and Apple Silicon targets) and RISC-V requires regeneration of FunctionID databases for each new target triple. The demangling and detection modules are architecture-independent and would transfer without modification.

Regarding adversarial robustness, the detection heuristics can be defeated by deliberate obfuscation. A malware author could strip Rust-specific strings, remove the .rustc section, modify the entry point structure, or apply binary-level obfuscation to evade identification. Our approach

prioritizes practical utility for the common case — standard toolchain compilation — over resilience to targeted evasion.

7.3. Implications for Malware Analysis

The growing adoption of Rust by malware authors creates a feedback loop: as Rust malware becomes more prevalent, the tooling gap becomes more consequential, and as tooling improves, the advantage malware authors gain from Rust compilation diminishes. Our plugin contributes to closing this gap on the Ghidra side of the ecosystem.

For incident responders and malware analysts, the practical impact is significant. The ability to quickly separate 127 application functions from 8,000+ library functions transforms Rust binary analysis from an intractable multi-day effort to a manageable task comparable in scope to analyzing a dynamically linked C binary. Combined with YARA rules for detecting Rust compiler artifacts and network traffic analysis for identifying command-and-control patterns, the plugin contributes to a practical Rust malware analysis workflow.

8. Conclusion

This paper has presented a method and its implementation as a Ghidra plugin for automatically handling Rust executables during reverse engineering analysis. The plugin addresses the primary challenges of Rust binary analysis through four integrated components: multi-heuristic binary detection, v0 symbol demangling with legacy format fallback, FunctionID-based standard library identification spanning 22 compiler versions, and pattern-based reconstruction of Rust-specific data types including Option, Result, string slices, and Vec.

Evaluation on a corpus of 35 binaries — including open-source utilities, CTF challenges, and real-world malware samples — demonstrates 100% detection accuracy, 96.8% demangling coverage, 74.2% standard library function identification in stripped binaries, and a statistically significant improvement in decompilation readability (2.2 to 3.9 on a 5-point scale) with a modest 18.5% analysis time overhead.

The work is positioned within a growing ecosystem of Rust reverse engineering tools and addresses a specific gap: the absence of a comprehensive, automated Rust analysis solution for the Ghidra platform. As Rust continues to grow in both legitimate software development and malware creation, tools that reduce the barrier to Rust binary analysis become increasingly critical for the cybersecurity community.

Future work will focus on: (1) integrating binary similarity matching (via Ghidra's BSim or a lightweight alternative) for version-tolerant function identification; (2) extending architectural support to AArch64 and RISC-V; (3) developing more robust enum layout reconstruction based on compiler layout algorithms; (4) exploring integration with ReOxide for decompiler-level Rust syntax emission; and (5) contributing upstream improvements to Ghidra's native Rust support through engagement with the open-source community.

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